

Definable classes in the weak Turing degrees

Josiah Jacobsen-Grocott

Nanyang Technological University

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- 1 Reducibilities on partial functions
- 2 Cone theorem
- 3 Semi-computable degrees
- 4 K-pairs
- 5 Definability results

There are several natural ways one could extend the usual definition “partial computable with oracle f ” to allow for cases where f itself is a partial function. Different, sometimes equivalent, extensions have been suggested.

One of the more systematic studies of the different notions of partial degrees was carried out by Sasso, and we will be building on his work here. This is joint work with Liling Ko and Keng Meng Ng.

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Computation with partial oracles

Definition (Turing functionals)

We define $\Phi_e(f)(n)$ to be the result of the e th program on input n and oracle f , i.e. the program is has the ability to query the values of $f(m)$.

How can we extend this to allow for a partial oracle?

Definition (Sasso)

We define $\Phi_i^{ST}(f)(n)$ to be the result of the i th program on input n and partial oracle f , with the understanding that if the program asks for the value of $f(m)$ for some m where, $f(m) \uparrow$ then $\Phi_i^{ST}(f)(n) \uparrow$.

With nondeterministic programs

Definition (Enumeration operators)

We define $\Phi_i^e(f)(n) = k$ if k is the output of *some* run the i th *nondeterministic* program on input n and partial oracle f , with the understanding that if the program asks for the value of $f(m)$ for some m where, $f(m) \uparrow$ then $\Phi_i^{ST}(f)(n) \uparrow$.

Observation

We have that $\text{graph}(g)$ is enumeration reducible ($\leq_e$) in the usual sense to $\text{graph}(f)$ if and only if there is some i such that $g = \Phi_i^e(f)$.

We can think of Φ_i^e as a c.e. set of axioms of the form (n, m, F) where F is a finite partial function represented by its canonical index and $(n, m, F) \in \Phi_i^e$ means that if $F \subseteq f$ then $\Phi_e^f(n) = m$.

Observation

It is possible that different runs of a nondeterministic program may produce different results, so it can happen that $\Phi_i^e(f)$ is multi valued.

Definition (Weak Turing operators)

We call a c.e. set of axioms W *consistent* if when ever $(n, m, F), (n, k, G) \in W$ then either $m = k$ or $F \cup G$ is not single valued. We define $\Phi_i^{WT}(f)(n) = k$ if the i th consistent c.e. set of axioms contains an axiom of the form (n, k, F) where $F \subseteq f$.

Degree structure

Definition

For $* \in \{ST, WT, e\}$ we can define $g \leq_* f$ if there exists i such that $g = \Phi_i^*(f)$. $\leq_*$ is a pre-order and taking equivalence classes gives us a degree structure $\mathcal{D}_*$.

Theorem (Sasso)

- We have that $f \leq_{ST} g \implies f \leq_{WT} g \implies f \leq_e g$.
- If f and g are total functions then
$$f \leq_{ST} g \iff f \leq_{WT} g \iff f \leq_e g \iff f \leq_T g.$$

Definition (Total degrees)

For $* \in \{ST, WT, e\}$ we can define $\mathcal{T}_* = \{\deg_*(f) : f \text{ is total}\}$.

We have $\mathcal{T}_* \cong \mathcal{D}_T$.

Definition (Sasso)

For a set $A \subseteq \omega$ we define the *characteristic* function of A :

$$C_A(n) = \begin{cases} 1 & n \in A \\ 0 & n \notin A \end{cases}$$

and the *semi*-characteristic function of A :

$$S_A(n) = \begin{cases} 0 & n \in A \\ \uparrow & n \notin A \end{cases}$$

$\mathcal{T}_* = \{\deg_*(C_A) : A \subseteq \mathbb{N}\}$ and $\mathcal{S}_* = \{\deg_*(S_A) : A \subseteq \mathbb{N}\}$

Theorem (Sasso)

$\mathcal{S}_e = \mathcal{D}_e$ and $\mathcal{S}_{WT} \cong \mathcal{D}_e$ is an initial segment of $\mathcal{D}_{WT}$.

Jump and semi-jump

Definition (Sasso)

For a partial function f define $K_f = \{\langle i, n, m \rangle : \Phi_i(f)(n) = m\}$. Using K_f we can define the *jump* of f to be $f' = C_{K_f}$ and the *semi-jump* of f to be $f^0 = S_{K_f}$.

Proposition (Sasso)

The following are properties of the jump and semi-jump:

- ① *We have that $f \leq_{WT} g \implies f' \leq_{WT} g' \wedge f^0 \leq_{WT} g^0$.*
- ② *For all A , $C'_A \equiv_{WT} C_{A'}$.*
- ③ *For any $S_A \leq_{WT} f$ we have: $S_A \leq_{WT} f^0 \equiv_{WT} (f^0)^0 \leq_{WT} f <_{WT} f'$.*
- ④ *For all f, g , $(f \oplus g)^0 \equiv_{WT} f^0 \oplus g^0$.*
- ⑤ *The semi-jump induces an upper semi-lattice embedding of $\mathcal{T}_{WT}$ into S_{WT} and under the isomorphism $I : \mathcal{D}_e \rightarrow S_{WT}$ we have that $I(\mathcal{T}_e) = (\mathcal{T}_{WT})^0$.*

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The cone theorem and the quotient structure

Theorem (Cone theorem, Sasso)

If $f \leq_{WT} g \leq_{WT} f \oplus S_A$ then there is $B \subseteq \omega$ such that $g \equiv_{WT} f \oplus S_B$.

Definition

- For partial functions f, g we define $f \leq_S^{WT} g$ if there is $A \subseteq \omega$ such that $f \leq_{WT} g \oplus S_A$.
- Equivalently we can define $f \leq_S^{WT} g$ if there is an WT -operator such that $f \subseteq \Phi_i^{WT}(g)$.
- We can think of D_S^{WT} as $\mathcal{D}_{WT}/\mathcal{S}_{WT}$.

Theorem

For partial functions f, g we have $f \leq_{WT} g$ if and only if $f^0 \leq_{WT} g^0$ and $f \leq_S^{WT} g$ if and only if $f \leq_e g$ and $f \leq_S^{WT} g$

The quotient structure

Corollary

For each $\mathcal{D}_S^{WT}$ -degree a we have that the semi-jump induces an embedding of $\{\deg_{WT}(f) : f \in a\}$ into $\mathcal{S}_{WT}$ as an upwards closed class of degrees. Furthermore, if $a \leq_S b$ then $a^0 \subseteq b^0$.

Corollary

$\mathcal{D}_{WT}$ is isomorphic to $\{(a, b) \in \mathcal{S}_{WT} \times \mathcal{D}_S^{WT} : a \in b^0\}$ under pairwise reducibility.

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Motivating question

Question

What is the structure of $\mathcal{D}_S^{WT}$. Is it the same as $\mathcal{D}_e$ or $\mathcal{D}_T$?

Theorem (Sasso)

$\mathcal{D}_{WT} \neq \mathcal{S}_{WT} \oplus \mathcal{T}_{WT}$. Hence $\mathcal{D}_S^{WT}$ is not just $\mathcal{D}_T$.

Definition

We say that a degree $a \in \mathcal{D}_{WT} \setminus \{0\}$ is *$\mathcal{S}$ -quasi-minimal* if for all $b \in \mathcal{D}_{WT}$, $b^0 = a^0 \implies a = b$.

Definition

For a real $r \in \mathbb{R}$, we define $N_r = \{n : q_n < r\}$ where $\{q_n\}_{n \in \omega}$ is the standard enumeration of the rationals. We call a degree $a \in \mathcal{D}_e$ *semi-computable* if $N_r \in (a)$ for some r that is neither left c.e. or right c.e. We can extend this definition to $\mathcal{S}_{WT}$ by identifying N_r with $S_r := S_{N_r}$.

Theorem

If $a \in \mathcal{S}_{WT}$ is semi-computable and $f^0 \leq_{WT} a$ then $f \leq_{WT} a$.

Theorem

If $a \in \mathcal{S}_{WT}$ is semi-computable and $f^0 \leq_{WT} a$ then $f \leq_{WT} a$.

Proof.

Let $S_r \in a$ be semi-computable and suppose that $f^0 \leq_{WT} S_r$. Then there is an e -operator Φ_i^e such that $f = \Phi_i^e(S_r)$. Consider the set $W = \{n : \exists m \neq k[(n, m), (n, k) \in \Phi_i^e(S_{q_n})]\}$. Since W is c.e. $t = \inf(W)$ is a right c.e. real and so $r \neq t$. Since $\Phi_i^e(S_r)$ is a function, it must be that $t \geq r$, and so $t > r$. Let q be a rational such that $r < q < t$.

Consider the operator

$\Phi = \Phi_i^e \setminus \{(n, k, \sigma) : \max(\text{dom}(\sigma)) \geq_{\mathbb{Q}} q \vee \text{range}(\sigma) \not\subseteq \{0\}\}$. This is a c.e. operator, since we removed a computable set, and for all g we have $\Phi(g) \subseteq \Phi(S_q)$. Since $q < t$ this is always a function, and $f = \Phi(S_r)$, so we have that $f \leq_{WT} S_r$. □

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Set definition of $\mathcal{K}$ -pairs

Definition (Kalimullin '03)

A and B are a Kalimullin pair ($\mathcal{K}$ -pair) if there is a c.e. set $W \subseteq \omega^2$ such that $A \times B \subseteq W$ and $\overline{A} \times \overline{B} \subseteq \overline{W}$. A $\mathcal{K}$ -pair is called *trivial* if one of A, B is c.e.

Degree definition of $\mathcal{K}$ -pairs

Definition (Minimal pair property)

For a pair of degrees $a, b \in \mathcal{D}$ we say they are a $\mathcal{K}$ -pair if for all $c \in \mathcal{D}$ we have that $c = (a \vee c) \wedge (b \vee c)$. If neither a or b is 0 then we call them a *nontrivial* $\mathcal{K}$ -pair

If, for all $x \geq a$ and $y \geq b$, $\{x, y\}$ is a $\mathcal{K}$ -pair implies $x = a$ and $y = b$, then we call $\{a, b\}$ a *maximal* $\mathcal{K}$ -pair.

Theorem (Kalimullin)

For the enumeration degrees, the two above definitions are equivalent.

Corollary

If $\{a, b\}$ are a $\mathcal{K}$ -pair, and $c \leq b$, then $\{a, c\}$ is also a $\mathcal{K}$ -pair.

Definability results in $\mathcal{D}_e$

$\mathcal{K}$ -pairs have been used to prove that many classes are definable in the enumeration degrees.

Theorem (Kalimullin)

The jump function is definable in the enumeration degrees.

Theorem (Ganchev, Soskova)

If $\{a, b\}$ are a nontrivial $\mathcal{K}$ -pair in $\mathcal{D}_e$ then there is a pair of semi-computable degrees $\{x, y\}$ that form a maximal $\mathcal{K}$ -pair and have $a \leq_e x$ and $b \leq_e y$. Furthermore, the semi-computable degrees can be characterized as the degrees of halves of maximal $\mathcal{K}$ -pairs

Theorem (Ganchev, Soskova)

The total e -degrees can be defined as the joins of maximal $\mathcal{K}$ -pairs.

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$\mathcal{K}$ -pairs in $\mathcal{S}_{WT}$ are $\mathcal{K}$ -pairs in $\mathcal{D}_{WT}$

Lemma

If $\{a, b\}$ are a $\mathcal{K}$ -pair in $\mathcal{S}_{WT}$ then they are also a $\mathcal{K}$ -pair in $\mathcal{D}_{WT}$.

Proof.

Consider some $c \in \mathcal{D}_{WT}$, and suppose that $d \leq_{WT} a \vee c, b \vee c$. Then we have that $d^0 \leq_{WT} a \vee c^0, b \vee c^0$. Since $a \vee c, b \vee c \leq_S^{WT} c$ we have that $d \leq_S^{WT} c$. So by Theorem 14, we have that $d \leq_{WT} c$. $\square$

$\mathcal{K}$ -pairs in $\mathcal{D}_{WT}$ are $\mathcal{K}$ -pairs in $\mathcal{S}_{WT}$

Lemma

If $\{a, b\}$ are a $\mathcal{K}$ -pair in $\mathcal{D}_{WT}$ then $a, b \in \mathcal{S}_{WT}$.

Proof.

By Prop 26 we have that $\{a^0, b^0\}$ is also a $\mathcal{K}$ -pair. By Theorem ?? we can take $\{x, y\}$ to be a pair of semi-computable degrees that bound $\{a^0, b^0\}$. By Theorem 22 $\{x, y\}$ also bound $\{a, b\}$, and so $x, b \in \mathcal{S}_{WT}$. $\square$

This allows us to define $\mathcal{S}_{WT}$ in $\mathcal{D}_{WT}$ as the class of degrees that are below the join of some $\mathcal{K}$ -pair.

proof on board

Table of results

The class or relation	new shorthand	The first order formula
$\mathcal{K}$ -pairs		$(\forall x)[x = (a \vee x) \wedge (b \vee x)]$
Non-trivial $\mathcal{K}$ -pairs	$K(a, b)$	$(\forall x)[x = (a \vee x) \wedge (b \vee x)] \wedge (a \neq 0) \wedge (b \neq 0)$
Maximal $\mathcal{K}$ -pairs	$M(a, b)$	$K(a, b) \wedge (\forall x, y)[K(x, y) \wedge (a \leq x) \wedge (b \leq y) \rightarrow (x = a) \wedge (y = b)]$
$\mathcal{S}_{WT}$	$\mathcal{S}(a)$	$(\exists x, y)[K(x, y) \wedge (a \leq (x \vee y))]$
Semi-jump	$b = a^0$	$(b \leq a) \wedge (b = b^0) \wedge (\forall x)[(x \leq a) \wedge (x = x^0) \rightarrow (x \leq b)]$
Semi-jump of $\mathcal{T}_{WT}$	$\mathcal{T}^0(a)$	$a = 0 \vee (\exists x, y)[M(x, y) \wedge (a = (x \vee y))]$
$\mathcal{T}_{WT}$	$\mathcal{T}(a)$	$\mathcal{T}^0(a^0) \wedge (a = \sup\{x : x^0 = a^0\})$
Jump on $\mathcal{S}_{WT} \setminus \{0\}$	$J^e(a)$	$J^e(a) = \sup\{x \vee y : K(x, y) \wedge (x \leq a)\}$
$0'$	$0'$	$0' = \inf\{x : \mathcal{T}(x) \wedge \exists y[J^e(y) = x^0]\}$
Jump	$b = a'$	$(b = 0' \wedge a = 0) \vee (\mathcal{T}(b) \wedge b^0 = J^e(a^0))$

Table: Definable relations and their defining formula

- Can these results be used to define classes in $\mathcal{D}_{ST}$?
- What is the structure of $\mathcal{D}_S^{WT}$? Does a version of Selman's theorem hold?

Thank you

Thank You